

Complex Analysis – 8

Гармонические функции и интеграл Пуассона

Домашнее задание

1 сентября 2026 года

Теоретический минимум. Harmonic functions and harmonic conjugates; the mean value property and the maximum principle; the Poisson kernel and the Dirichlet problem in the disc and in the half-plane; Harnack's inequality and Harnack's principle; Liouville-type theorems for harmonic functions; subharmonic functions and the subharmonicity of $\log |f|$.

1. Solve the Dirichlet problem in the unit disc through the Poisson kernel and prove uniqueness of the solution; verify that the kernel is an approximate identity.
2. (S.T. Yau CSMC 2010) Find a harmonic function on the right half-plane whose limit is 1 at every point of the positive imaginary half-axis and -1 at every point of the negative imaginary half-axis.
3. (S.T. Yau CSMC 2018) Prove that if a sequence of harmonic functions on the open disc converges uniformly on compact subsets, then the limit is harmonic.
4. (S.T. Yau CSMC 2018) Prove that a positive harmonic function u on $\mathbb{R}^n \setminus \{0\}$, $n \geq 2$, has the form $u(x) = a + b|x|^{2-n}$ with constants $a, b \geq 0$.
5. (S.T. Yau CSMC 2019) Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be harmonic with $\lim_{|x| \rightarrow \infty} \frac{|f(x)|}{\ln |x|} = 0$. Prove or disprove that f is constant.
6. (S.T. Yau CSMC 2015, team) Let $\{u_n\}$ be harmonic on Ω with $\|u_n\|_{L^2(\Omega)} \leq M$ for a constant M independent of n . Prove that some subsequence converges in L^2 on compact subsets.

7. Prove Harnack's inequality for the disc and deduce Harnack's principle: an increasing sequence of harmonic functions either converges locally uniformly to a harmonic function or tends to $+\infty$ everywhere.
8. Prove the reflection principle for harmonic functions: a function harmonic in the upper half-disc, continuous up to the diameter and vanishing there, extends harmonically by $u(\bar{z}) = -u(z)$; deduce the Schwarz reflection principle for holomorphic functions.
9. Prove that every harmonic function on the annulus $\{r < |z| < R\}$ is of the form $a \log |z| + \operatorname{Re} g(z)$ with g holomorphic, and characterise those which possess a single-valued harmonic conjugate.
10. Prove that $\log |f|$ is subharmonic for holomorphic f , state the maximum principle for subharmonic functions, and deduce Hadamard's three-circles theorem from it.